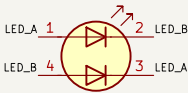


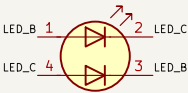
CELL LED BOARD (remote, top of the stock tube)

S1  
KPHBM-2012SURKCGKC



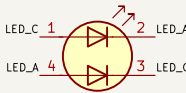
Cell 1: A + = red, B + = green

S2  
KPHBM-2012SURKCGKC



Cell 2: B + = red, C + = green

S3  
KPHBM-2012SURKCGKC



Cell 3: C + = red, A + = green

H1  
M2



H1 / H2: M2 mounting holes, one on each short side of the 20 x 10 mm board

Wire pads A / B / C: thin wires to the charger board pads TP13 (A), TP14 (B), TP15 (C).  
Charger side drives PA1 = A, PA3 = B, PA0 = C through 100R each (R27-R29) and scans one line + per ms.  
Cell 1 LED between A-B, cell 2 between B-C, cell 3 between C-A (the same order as the firmware).  
LED: Kingbright KPHBM-2012SURKCGKC (LCSC C598341), 2.0 x 1.25 mm, red 1.95 V / yellow-green 2.1 V.  
Datasheet pinout: red = anode 1, cathode 2; green = anode 4, cathode 3 -> pins 1+3 to the first line, pins 2+4 to the second line. Two LEDs in series need ~4 V > 3.3 V, so no LED lights by accident.

No parts but the LEDs: the 100R resistors are on the charger board (R27-R29)  
3 red/green LEDs, one per cell, charlieplexed from the charger pads A/B/C

Sheet: /  
File: led\_board.kicad\_sch

Title: Cell LED board

Size: A4 Date: 2026-10-02

KiCad E.D.A. 10.0.1

Rev: A  
Id: 1/1